

## Multiple Choice (10 marks)

1. The rotational spectrum of HF has lines  $41.9\text{ cm}^{-1}$  apart. Calculate the moment of inertia and the bond length.

2.  $1.256 \times 10^{-46}\text{ kg m}^2$  and  $9.100 \times 10^{-2}\text{ nm}$

2.  $1.336 \times 10^{-47}\text{ kg m}^2$  and  $8.250 \times 10^{-2}\text{ nm}$

2.  $1.336 \times 10^{-46}\text{ kg m}^2$  and  $9.196 \times 10^{-1}\text{ nm}$

2.  $1.410 \times 10^{-47}\text{ kg m}^2$  and  $9.250 \times 10^{-2}\text{ nm}$

2.  $1.256 \times 10^{-47}\text{ kg m}^2$  and  $8.196 \times 10^{-2}\text{ nm}$

Calculate the energy of a photon for a wavelength of 100 pm (about one atomic diameter). \$

(a)  $4 \times 10^{-15}\text{ J}$

(a)  $1.2 \times 10^{-15}\text{ J}$

(a)  $5 \times 10^{-15}\text{ J}$

(a)  $3.5 \times 10^{-15}\text{ J}$

(a)  $2 \times 10^{-15}\text{ J}$

Less than 1/1000 of the mass of any atom is contributed by

(a) the protons, electrons and neutrons

(b) the electrons and protons

(c) the nucleus

(d) the protons

(e) the neutrons

What is the electromotive force (emf) of a system when K, the equilibrium constant for the reaction = Q, the concentration quotient?

(a) -2

(b) 1

(c) -0.5

(d) 0.5

(e) 2

What is the heat capacity of uranium? Its atomic heat is 26 J/mole and its heat capacity is 238 g/mole.

- (a) 0.11 J/g
- (a) 0.20 J/g
- (a) 0.13 J/g
- (a) 0.18 J/g
- (a) 0.30 J/g

## True / False (10 marks)

*Answer True or False*

- 6. True or False: The oxidation number of chlorine in the perchlorate ion,  $\text{ClO}_4^-$ , is +5.
- 7. True or False: The strongest base in liquid ammonia is ammonia itself,  $\text{NH}_3$ .
- 8. True or False: An atom of  $^{52}\text{Cr}$  contains 28 protons, 24 neutrons, and 24 electrons.
- 9. True or False: The compounds formed from Sn and F, P and H, and Si and O are  $\text{SnF}_4$ ,  $\text{PH}_3$ , and  $\text{Si}_2\text{O}_5$ , respectively.
- 10. True or False: A  $T_2$  value of 15 ms results in a linewidth of approximately 52.8 Hz in NMR spectroscopy.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

- 11. Discuss the reasons why the  $^1\text{H}$  NMR spectrum of  $\text{CHCl}_3$  appears as a singlet, focusing on the implications of the electric quadrupole moments of  $^{35}\text{Cl}$  and  $^{37}\text{Cl}$ .
- 12. Given an unknown element X with an atomic weight of 210.197 amu and two isotopes,  $^{210}\text{X}$  and  $^{212}\text{X}$  with respective masses of 209.64 and 211.66 amu, calculate the relative abundance of each isotope.

*End of Paper*